AMS 526: Numerical Linear Algebra for Computational and Data Sciences

Lecture 4: Singular Value Decomposition

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Outline

1. **Singular Value Decomposition (NLA §4–5)**

Singular Value Decomposition (SVD)

- Given $A \in \mathbb{R}^{m\times n}$, its *SVD* is
$$A = U\Sigma V^{\mathsf{T}},$$
where $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are orthogonal, and $\Sigma \in \mathbb{R}^{m\times n}$ is diagonal.
- If $A \in \mathbb{C}^{m\times n}$, then its SVD is $A = U\Sigma V^{\mathsf{H}}$, where $U \in \mathbb{C}^{m\times m}$ and $V \in \mathbb{C}^{n\times n}$ are unitary, and $\Sigma \in \mathbb{R}^{m\times n}$ is diagonal.
- *Singular values* are the diagonal entries of Σ, denoted by $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{\min\{m,n\}} \geq 0$.
- *Left singular vectors* of A are the column vectors of U.
- *Right singular vectors* of A are the column vectors of V.
- We have $Av_j = \sigma_j u_j$ for $1 \leq j \leq \min\{m, n\}$.
- SVD plays a prominent role in data analysis and matrix analysis.

Geometric Observation

- Given a unit hypersphere S in $\mathbb{R}^n$, let AS denote the shape after transformation, which is a *hyperellipsoid* in $\mathbb{R}^m$.
- Columns in V are the preimages of the principal semiaxes of the hyperellipsoid AS.
- *Singular values* correspond to the lengths of the principal semiaxes of the hyperellipsoid.
- *Left singular vectors* are parallel to the principal semiaxes of AS.
- *Right singular vectors* are preimages of those semiaxes.

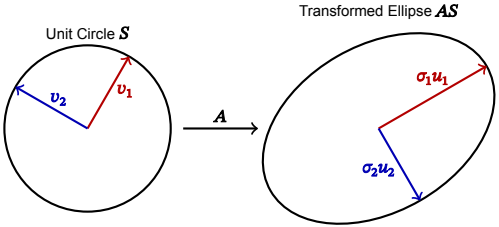

Figure: Geometric interpretation of SVD transforming a unit circle S into an ellipse AS.

Two Different Types of SVD

- **Full SVD**: For $A \in \mathbb{R}^{m \times n}$, we have $U \in \mathbb{R}^{m \times m}$, $\Sigma \in \mathbb{R}^{m \times n}$, and $V \in \mathbb{R}^{n \times n}$ such that

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- **Thin SVD** (or **Reduced SVD**): Assuming $m \geq n$, we have $\hat{U} \in \mathbb{R}^{m\times n}$ and $\hat{\Sigma} \in \mathbb{R}^{n\times n}$ such that

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$$A = \hat{U}\hat{\Sigma}V^\top.$$

- Furthermore, notice that

$$A = \sum_{i=1}^{\min\{m,n\}} \sigma_i u_i v_i^\top.$$

  Thus, we can keep only entries of U and V corresponding to nonzero σ_i.

Existence of SVD

Theorem (Existence)

Every matrix $A \in \mathbb{R}^{m \times n}$ has an SVD.

Proof

Let $\sigma_1 = \|A\|_2$. There exists $v_1 \in \mathbb{R}^n$ with $\|v_1\|_2 = 1$ and $\|Av_1\|_2 = \sigma_1$. Let U_1 and V_1 be orthogonal matrices whose first columns are $u_1 = Av_1/\sigma_1$ (or any unit-length vector if $\sigma_1 = 0$) and v_1, respectively. Note that (using block-matrix notation)

$$U_1^\top A V_1 = S = \begin{bmatrix} \sigma_1 & \omega^\top \\ 0 & B \end{bmatrix}. \tag{1}$$

(Key remaining questions: What is ω, and how to deal with B?)

Existence of SVD Cont'd

Proof (Proof Cont'd)

Furthermore, $\omega = 0$ because $\|S\|_2 = \sigma_1$, and

$$\left\| \begin{bmatrix} \sigma_1 & \omega^\top \\ 0 & B \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \omega \end{bmatrix} \right\|_2 \geq \sigma_1^2 + \omega^\top \omega = \sqrt{\sigma_1^2 + \omega^\top \omega} \left\| \begin{bmatrix} \sigma_1 \\ \omega \end{bmatrix} \right\|_2,$$

implying that $\sigma_1 \geq \sqrt{\sigma_1^2 + \omega^\top \omega}$ and thus $\omega = 0$.

We then prove by induction using (1) from the previous slide. If $m = 1$ or $n = 1$, then B is empty, and we have $A = U_1 S V_1^\top$. Otherwise, suppose $B = U_2 \Sigma_2 V_2^\top$. Then

$$A = \underbrace{U_1 \begin{bmatrix} 1 & 0^\top \\ 0 & U_2 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \sigma_1 & 0^\top \\ 0 & \Sigma_2 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 1 & 0^\top \\ 0 & V_2^\top \end{bmatrix} V_1^\top}_{V^\top},$$

where U and V are orthogonal, and $\sigma_1 \geq \sigma_i$ for any σ_i in Σ_2.

Uniqueness of SVD

Theorem (Uniqueness)

The singular values $\{\sigma_j\}$ are uniquely determined. If A is square and the σ_j are distinct, the left and right singular vectors are uniquely determined **up to signs**.

Geometric argument: If the lengths of the semiaxes of a hyperellipsoid are distinct, then the orientations of the semiaxes are determined up to signs.

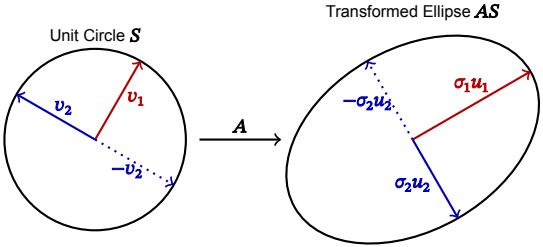

Figure: Geometric illustration of the uniqueness of SVD up to signs.

Uniqueness of SVD Cont'd

Algebraic argument: The proof can be done by induction. Consider the case where the σ_j are distinct. The 2-norm is unique, so σ_1 is unique. If v_1 is not unique up to sign, then the orthonormal bases of these vectors are right singular vectors of A, implying that σ_1 is not a simple singular value.

Once the first triplet $\sigma_1, u_1,$ and v_1 is determined, the remainder of SVD follows from the subspace orthogonal to v_1. Because v_1 is unique up to sign, the orthogonal subspace is uniquely defined. The rest of the SVD can then be uniquely determined by induction.

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- **Discussion**: What if we change the sign of a singular vector?
- **Discussion**: What if σ_i is not distinct?

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- Eigenvalue decomposition of a nondefective matrix A is $A = X\Lambda X^{-1}$.
- Differences between SVD and eigenvalue decomposition:
  - Not every matrix has an eigenvalue decomposition, but every matrix has a singular value decomposition.
  - Eigenvalues may not always be real numbers, but singular values are always non-negative real numbers.
  - Eigenvectors are not always orthogonal to each other (they are orthogonal for symmetric matrices), but left (or right) singular vectors are orthogonal to each other.

SVD vs. Eigenvalue Decomposition Cont'd

- Similarities:
  - Singular values of $\boldsymbol{A}$ are the square roots of the eigenvalues of $\boldsymbol{A}\boldsymbol{A}^\top$ and $\boldsymbol{A}^\top\boldsymbol{A}$, and their eigenvectors are left and right singular vectors, respectively.
  - Singular values of Hermitian matrices are the absolute values of eigenvalues, and there exists an SVD such that the eigenvectors are the singular vectors.
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  - Singular values of Hermitian matrices are the absolute values of eigenvalues, and there exists an SVD such that the eigenvectors are the singular vectors.
  - This relationship can be used to compute singular values by hand.
- **Discussion**: Are the eigenvalues and eigenvectors of $\boldsymbol{A}\boldsymbol{A}^\top$ unique?

Matrix Properties via SVD

- Let r be the number of nonzero singular values σ_i of $A \in \mathbb{R}^{m \times n}$.
  - $\text{rank}(A)$ is r.
  - $\text{range}(A) = \text{span}\{u_1, u_2, \dots, u_r\}$.
  - $\text{null}(A) = \text{span}\{v_{r+1}, v_{r+2}, \dots, v_n\}$.
- 2-norm and Frobenius norm:
  - $\|A\|_2 = \sigma_1$ and $\|A\|_F = \sqrt{\sum_i \sigma_i^2}$.
- Determinant of matrix:
  - For $A \in \mathbb{R}^{m \times m}$, $|\det(A)| = \prod_{i=1}^{m} \sigma_i$.

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  - For $A \in \mathbb{R}^{m \times m}$, $|\det(A)| = \prod_{i=1}^{m} \sigma_i$.
- However, SVD may not be the most efficient way to solve problems.
- Algorithms for SVD are similar to those for eigenvalue decomposition, and we will discuss them later.